

data_collection.ipynb

File Edit View Insert Runtime Tools Help

Commands Code Text Run all

RAM Disk

Files

.. EduVision_DV O1_raw_datasets QS_2025 QS World University ... THE_2024 TIMES_WorldUniversi... WUR_2023 World University Ran... WorldBank_Education edstats-csv-zip-32-... EdStatsCountry-S... EdStatsCountry.csv EdStatsData.csv

Disk 86.83 GB available

[2] 20s

EduVision_DV - data_collection.py Downloads ONLY the four approved raw datasets required by EduVision_DV. Datasets: 1. QS World University Rankings 2025 2. Times Higher Education (THE) World University Rankings 2024 3. World University Rankings 2023 (WUR 2023) 4. World Bank Education Statistics / EdStats Designed for Google Colab. No cleaning, renaming, merging, fuzzy matching, KPI engineering, or transformation is performed here. Raw downloaded files are preserved. Sources selected according to the EduVision_DV project guide: - QS 2025: melissamonfared/qs-world-university-rankings-2025 - THE 2024: ddosad/timesworlduniversityrankings2024 - WUR 2023: alitaqi000/world-university-rankings-2023 - World Bank Education: theworldbank/education-statistics ##### from pathlib import Path import sys import subprocess import shutil import json from datetime import datetime # ----- # 1. Install the ONLY external package required for downloading # ----- def install_dependencies(): try: import kagglehub # noqa: F401 except ImportError: subprocess.check_call([sys.executable, "-m", "pip", "install", "-q", "kagglehub"]) install_dependencies() import kagglehub # ----- # 2. Configuration # ----- BASE_DIR = Path.cwd() / "EduVision_DV" RAW_DIR = BASE_DIR / "01_raw_datasets" RAW_DIR.mkdir(parents=True, exist_ok=True) DATASETS = { "qs_2025": { "kaggle_id": "melissamonfared/qs-world-university-rankings-2025", "folder": "QS_2025", "expected_year": 2025, }, "the_2024": { "kaggle_id": "ddosad/timesworlduniversityrankings2024", "folder": "THE_2024", "expected_year": 2024, }, "wur_2023": { "kaggle_id": "alitaqi000/world-university-rankings-2023", "folder": "WUR_2023", "expected_year": 2023, }, "country_comparison": { "kaggle_id": "theworldbank/education-statistics", "folder": "WorldBank_Education", "expected_year": "country/year level", }, }, } # ----- # 3. Helper functions # ----- def copy_downloaded_dataset(dataset_key, source_dir, destination_dir): """Copy the complete downloaded dataset into the project's raw folder.""" destination_dir.mkdir(parents=True, exist_ok=True) source_dir = Path(source_dir) if not source_dir.exists(): raise FileNotFoundError(f"Downloaded dataset directory does not exist: {source_dir}") copied_files = [] # Copy every file recursively. This is intentional: # raw datasets must remain unchanged. for item in source_dir.rglob("*"): if item.is_file(): relative_path = item.relative_to(source_dir) target = destination_dir / relative_path target.parent.mkdir(parents=True, exist_ok=True) shutil.copy2(item, target) copied_files.append(str(target.relative_to(BASE_DIR))) if not copied_files: raise RuntimeError(f"No files were found inside the downloaded dataset for {dataset_key}.") return copied_files def write_manifest(results): """Write collection metadata without modifying the raw datasets.""" manifest = { "project": "EduVision_DV", "created_at": datetime.now().isoformat(timespec="seconds"), "purpose": "Raw dataset collection only", "datasets": results, } manifest_path = RAW_DIR / "data_collection_manifest.json" with open(manifest_path, "w", encoding="utf-8") as f: json.dump(manifest, f, indent=2, ensure_ascii=False) return manifest_path def download_one(dataset_key, config): print("\n" + "-" * 72) print(f"Downloading: {dataset_key}") print(f"Kaggle dataset: {config['kaggle_id']}") print("-" * 72) destination_dir = RAW_DIR / config["folder"] # Download from Kaggle using kagglehub. # For public datasets, this is the only dataset-download mechanism used. downloaded_path = kagglehub.dataset_download(config["kaggle_id"]) copied_files = copy_downloaded_dataset(dataset_key=dataset_key, source_dir=downloaded_path, destination_dir=destination_dir,) print(f"Downloaded successfully: {len(copied_files)} file(s)") print(f"Saved to: {destination_dir}") return { "dataset_key": dataset_key, "kaggle_id": config["kaggle_id"], "expected_year": config["expected year"], "destination": str(destination_dir), "files": copied_files, "status": "SUCCESS", } # ----- # 4. Main collection process # ----- def main(): print("\nEduVision_DV - Raw Dataset Collection") print("Google Colab compatible") print("Collection only: NO cleaning / NO merging / NO KPI engineering") print(f"Output directory: {BASE_DIR}\n") results = [] failures = [] for dataset_key, config in DATASETS.items(): try: result = download_one(dataset_key, config) results.append(result) except Exception as exc: print(f"\nFAILED: {dataset_key}") print(f"Reason: {exc}") failures.append({ "dataset_key": dataset_key, "kaggle_id": config["kaggle_id"], "status": "FAILED", "error": str(exc), }) manifest_path = write_manifest(results + failures) print("\n" + "-" * 72) print("COLLECTION SUMMARY") print("-" * 72) for result in results: print(f"[OK] {result['dataset_key']}: {result['kaggle_id']}") for failure in failures: print(f"[FAIL] {failure['dataset_key']}: {failure['kaggle_id']}") print(f"\nManifest: {manifest_path}") if failures: print("\nSome datasets failed to download. " "Check the error above and rerun the script.") return 1 print("\nAll 4 approved datasets downloaded successfully.") print(f"Raw datasets are preserved under: {RAW_DIR}") return 0 if __name__ == "__main__": raise SystemExit(main())

EduVision_DV - Raw Dataset Collection

Google Colab compatible

Collection only: NO cleaning / NO merging / NO KPI engineering

Output directory: /content/EduVision_DV

Downloading: qs_2025 Kaggle dataset: melissamonfared/qs-world-university-rankings-2025 ##### Using Colab cache for faster access to the 'qs-world-university-rankings-2025' dataset. Downloaded successfully: 1 file(s) Saved to: /content/EduVision_DV/01_raw_datasets/QS_2025 ##### Downloading: the_2024 Kaggle dataset: ddosad/timesworlduniversityrankings2024 ##### Using Colab cache for faster access to the 'timesworlduniversityrankings2024' dataset. Downloaded successfully: 1 file(s) Saved to: /content/EduVision_DV/01_raw_datasets/THE_2024 ##### Downloading: wur_2023 Kaggle dataset: alitaqi000/world-university-rankings-2023 ##### Using Colab cache for faster access to the 'world-university-rankings-2023' dataset. Downloaded successfully: 1 file(s) Saved to: /content/EduVision_DV/01_raw_datasets/WUR_2023 ##### Downloading: country_comparison Kaggle dataset: theworldbank/education-statistics ##### Downloading to /root/.cache/kagglehub/datasets/theworldbank/education-statistics/45.archive... 100%[]| 75.6M/75.6M [00:00<00:00, 164MB/s]Extracting files... Downloaded successfully: 6 file(s) Saved to: /content/EduVision_DV/01_raw_datasets/WorldBank_Education ##### COLLECTION SUMMARY ##### [OK] qs_2025: melissamonfared/qs-world-university-rankings-2025 [OK] the_2024: ddosad/timesworlduniversityrankings2024 [OK] wur_2023: alitaqi000/world-university-rankings-2023 [OK] country_comparison: theworldbank/education-statistics Manifest: /content/EduVision_DV/01_raw_datasets/data_collection_manifest.json All 4 approved datasets downloaded successfully. Raw datasets are preserved under: /content/EduVision_DV/01_raw_datasets An exception has occurred, use %tb to see the full traceback. SystemExit: 0

Next steps: Explain error

Colab paid products - Cancel contracts here

Variables Terminal 2:41 AM Python 3